Waipapa Taumata Rau
University of Auckland

Part IV Project #123
Department of Electrical, Computer, and Software Engineering

AI-Based Virtual Parliament

Cooperative debate and hidden adversarial strategies

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Real MPs can derail debate, distort claims and exploit procedure. AI multi-agent debate has been studied to improve reasoning [2]. We built a virtual NZ House of AI MPs, then privately tasked some with sabotage and tested Hansard grounding.

1 Parliament web browser interface

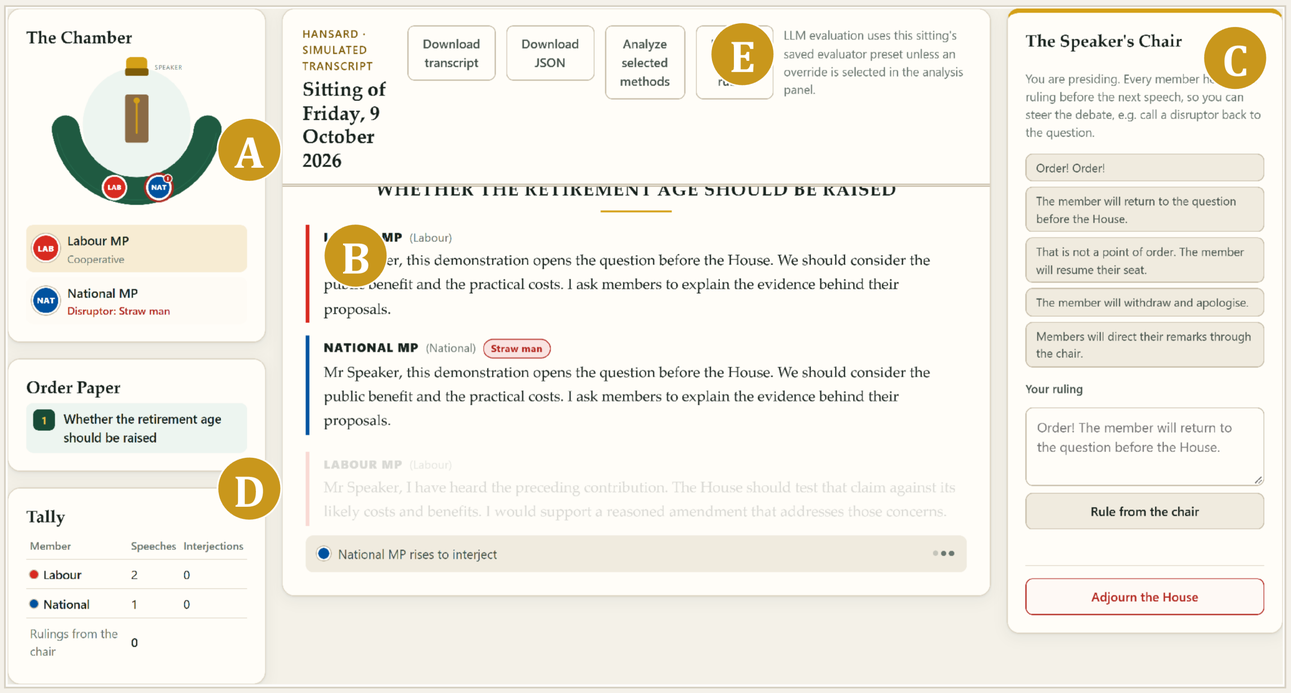


Fig. 1 A demonstration sitting with a National straw-man assignment.
Web browser, guided command, and terminal interfaces all connect to the debate engine backend.

A The Chamber
One MP represents each chosen party.
Speakers light up; adversarial roles appear in red.

B Live Hansard
Speeches and interjections stream into the live transcript.
Saved sittings can be replayed after a restart.

C The Speaker's Chair
Chair rulings enter the shared public record and reach every MP before the next speech.

D Order Paper and Tally
The agenda advances through topics in order.
Live totals track speeches, interjections and rulings.

E Exports and evaluation
Public JSON and text exports support local sentiment/stance analysis and separate rubric scores.

2 Cooperative and adversarial agents

At run start, the backend combines each MP's party profile and selected real Hansard excerpts [1] in a private system prompt, adding any assigned tactic. The scheduler takes turns sending that MP's prompt, public debate history and a turn cue to its model's chat completion endpoint. Each returned speech joins the public record for the next MP. Other agents cannot read the private prompts.

Real Hansard
500 reviewed excerpts
0/10/100 per party

→ **Private setup**
Frozen prompts
and hidden tactics

→ **MP agents**
Party personas
on cloud or local LLMs

→ **Debate engine**
Scheduled turns
and chat requests

→ **Saved evidence**
Public transcript
and separate reports

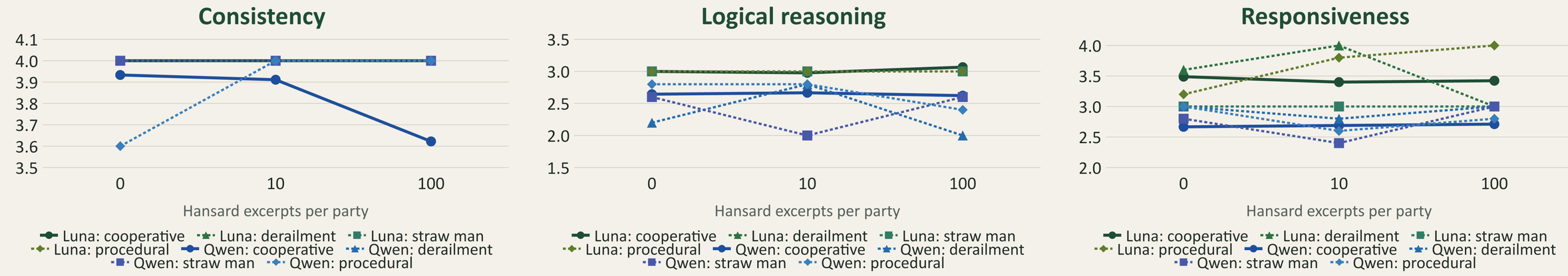
Topic derailment
Steer discussion away from the agenda without acknowledging it.

Straw man
Misrepresent an opponent's claim, then attack the distorted version.

Procedural manipulation
Exploit rules or points of order to obstruct substantive debate.

3 Grounding and adversarial testing

72 debates (1,800 speeches): **54 cooperative baselines** (three repeats), plus **18 adversarial runs** (one repeat, one assigned MP). GPT-6 Luna and Qwen3:8b debate retirement, nuclear power and a four-day workweek at 0/10/100 excerpts, with five parties, five rounds and no interruptions.



Vertical axes are truncated. Rubric scores range from 0 to 4.

Findings: Consistency stays high, but Qwen's cooperative score falls from 3.93 to 3.62. Luna's reasoning stays near 3.0; Qwen's strategy scores range from 2.0 to 2.8. Responsiveness varies by strategy: Luna's procedural score rises from 3.2 to 4.0. More Hansard gives no consistent gain.

Disruption: Qwen ACT scores 2/4 for disruptive rhetoric only at 0 excerpts; assigned Luna MPs score 0. Matched Qwen Green changes range from -1 to +1. Higher scores mean more disruption; assignment alone does not establish successful disruption.

Cooperative points average 9 debates (45 scores); each strategy point averages all five MPs in one debate. Assignments: retirement/NZ First/derailment, workweek/ACT/straw man, nuclear/Green/procedural. A consistent prompt to GPT-6 Luna generated the ratings. Topic, party and one assigned repeat limit comparison.

4 Outcomes and next steps

Platform: An AI debate engine with replaceable components and a browser-based frontend. Our study found no consistent score gain from extra Hansard.

Next: Repeat assigned conditions and rotate parties. Human reviewers should assess disruption, reasoning and concessions before wider conclusions.

Contributions: Albert Sun developed the frontend, live chamber, initial CLI and Hansard grounding. **Kieran Joe** developed the engine and architecture, and coordinated with the supervisor.

References

[1] Rauh & Schwalbach (2020). ParlSpeech V2.
[2] Du et al. (2024). Multiagent Debate.

Acknowledgements

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